Shivam Vipul Kapadia

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Education

University of Illinois Urbana-Champaign | *B.S. Computer Engineering, Semiconductor Minor* Expected Dec. 2028

- **Honors:** GPA 3.96/4.00, Dean's List (Fall 2025, Spring 2026), James Scholar, IEEE-HKN
- **Scholarships:** Grainger Engineering Achievement Scholarship, A.R. "Buck" Knight Scholarship
- **Coursework:** Analog Signal Processing, Electronics, Electricity and Magnetism, Computer Systems and Programming
- **Illinois ECE:** Course Assistant and Grader, ECE 110 Electronics; Grader, ECE 120 Computing
- **Department service:** ECESAC Academic Committee Member; Front Desk Representative

Skills

Embedded: Embedded C, STM32 HAL, STM32CubeIDE, Arduino C++, UART, I2C, SPI, CAN, GPIO interrupts
Hardware: Analog circuit design, 555 timer circuits, Digital counters, Motor control, Antenna evaluation, KiCAD, AutoCAD
Programming: C/C++, Python, LC-3 Assembly, Java, SQL, TypeScript, JavaScript, Bash
Software and tools: FastAPI, PostgreSQL, Python asyncio, MediaPipe, Docker Compose, Git, GitHub Actions, pytest

Projects

Hands-Free Dishwasher | *Analog Electronics, 555 Timer, ECE 145 Honors Design Laboratory* Jan. – May 2026

- A hands-free dishwashing appliance sized for a dorm room, built over a semester in the ECE 145 honors design lab
- Devised the wash cycle as cascaded **counters** clocked by a **555 timer**, so each stage hands off with no code
- Fabricated a compact, fully **analog** build around safety and part count, with **LEDs** showing each stage

Meter, LLM Cost Governance Proxy | *Python, FastAPI, PostgreSQL, asyncio* Aug. 2026

- A drop-in **proxy** in front of an AI team's **LLM** calls that meters spend and refuses any request over budget
- Owned the payments and treasury workstream on a four-person, three-day build: **2,000 lines**, **205 tests**
- Isolated blocking calls from the shared **asyncio event loop**, so one stalled call cannot freeze every request
- Diagnosed **four defects** disabling payments, one a **livelock** where checking a backoff timer kept resetting it

PalmSlash, Real-Time Hand Tracking | *JavaScript, MediaPipe, p5.js* June 2026

- A webcam browser game that tracks both hands in real time, running fully **client-side** with no server
- Optimized tracking to a steady **60 fps** by throttling **MediaPipe** inference to **22 Hz** and filtering between samples
- Established a fingertip speed **threshold** between frames, so jitter from a resting hand never counts as a swipe

Experience

Illinois EV Concept | *Embedded Systems Member* Sep. 2026 – Present

- A student team building an electric vehicle, whose embedded group is adding a radio **telemetry** link off the car
- Writing an **Embedded C** driver that links an **STM32WL** radio chip to the car's other boards for telemetry
- Planning an **SPI**-to-**CAN** bridge on an **MCP2515**, since the STM32WL has no CAN port to read the car's bus
- Researching **antennas** for the radio, comparing candidates and **range testing** the link on an STM32WL board
- Configured **STM32** boards in onboarding to read an **I2C** sensor on a **GPIO interrupt**, logged over **UART**

BeatBot, ASME Robotic Drummer Project | *Embedded Systems Engineer* Jan. – Apr. 2026

- A two-armed robotic drummer that plays programmed songs on a real kit, developed with an ASME team over a semester
- Programmed the **Arduino C++** that sequences and times every drum strike, as primary contributor on the ECE side
- Assembled the **brushless DC motor** striking arms and contributed to the **CAD** design of the mechanism they swing on

Revamp, Agentic AI Consulting at UIUC | *Software Engineer* Sep. 2026 – Present

- **Clearline**: a tool turning scanned billing PDFs into reviewed spreadsheets, a job of **6–9 hours** a week by hand
- Structured corrections as an **overlay**, so extracted values, **OCR** confidence and source pages are never overwritten
- Enforced **all-or-nothing** validation on **16** fields, backed by **59 tests** green on **Ubuntu** and **Windows**

Pandit Deendayal Energy University | *Machine Learning Research Assistant* Jan. 2024 – Feb. 2025

- **Paper 1, Springer 2025:** Compared three air quality forecasting methods across published results; combining them won
- **Paper 2, rainfall:** Surveyed **70 papers**, mapped contradictions, proposed phone-location flood alerts; first author

1Rivet | *Software Engineering Intern* May 2024

- Built the web screens a manufacturing client's customers used, hand-coding the whole front end in **HTML** and **CSS**
- Wrote the **SQL** queries that store and retrieve company records, and ran backup, recovery and purge routines